EXP:3	EDA-Data Cleaning

CODE:

```
import pandas as pd

import numpy as np

from sklearn.preprocessing import MinMaxScaler, StandardScaler


data = {

    'Gender': ['Male', 'Female', 'Male', 'Female', 'Male', 'Female', 'Male', 'Female', 'Male', 'Female'],

    'Married': ['Yes', 'No', 'Yes', 'Yes', 'No', 'Yes', 'No', 'Yes', 'Yes', 'No'],

    'Dependents': [1, 0, 2, 1, '3+', 0, 2, 1, 0, 3],

    'Education': ['Graduate', 'Graduate', 'Not Graduate', 'Graduate', 'Graduate', 'Not Graduate', 'Graduate', 'Graduate', 'Not Graduate', 'Graduate'],

    'Self_Employed': ['No', 'No', 'Yes', 'No', 'Yes', 'No', 'No', 'Yes', 'No', 'No'],

    'ApplicantIncome': [5000, 3000, 7000, 4500, 8000, 3500, 6000, 4000, 5500, 3200],

    'CoapplicantIncome': [1500, 0, 2000, 1000, 2500, 500, 1200, 0, 1800, 700],

    'LoanAmount': [150, 100, 200, 120, 250, 110, 180, 130, 160, 105],

    'Loan_Amount_Term': [360, 360, 360, 240, 360, 360, 180, 360, 240, 360],

    'Credit_History': [1, 1, 0, 1, 1, 1, 0, 1, 1, 1],

    'Property_Area': ['Urban', 'Rural', 'Semiurban', 'Urban', 'Rural', 'Semiurban', 'Urban', 'Rural', 'Semiurban', 'Urban']

}


df = pd.DataFrame(data)
```

```
df.loc[2, 'Gender'] = np.nan
df.loc[4, 'LoanAmount'] = np.nan
df.loc[7, 'Credit_History'] = np.nan

df = pd.concat([df, df.iloc[[0]]], ignore_index=True)

print("Initial Data Overview:")
print(df.info())

print("\nMissing Values:")
print(df.isnull().sum())

df['Gender'] = df['Gender'].fillna(df['Gender'].mode()[0])
df['Married'] = df['Married'].fillna(df['Married'].mode()[0])
df['Dependents'] = df['Dependents'].fillna(0)
df['Self_Employed'] =
df['Self_Employed'].fillna(df['Self_Employed'].mode()[0])
df['LoanAmount'] = df['LoanAmount'].fillna(df['LoanAmount'].median())
df['Loan_Amount_Term'] =
df['Loan_Amount_Term'].fillna(df['Loan_Amount_Term'].mode()[0])
df['Credit_History'] = df['Credit_History'].fillna(df['Credit_History'].mode()[0])

initial_rows = len(df)
df.drop_duplicates(inplace=True)

print("\nDuplicates Removed:", initial_rows - len(df))
```

```
df['Dependents'] = df['Dependents'].replace('3+', 3)
df['Dependents'] = pd.to_numeric(df['Dependents'],
errors='coerce').fillna(0).astype(int)

for col in ['Gender', 'Married', 'Education', 'Self_Employed', 'Property_Area']:
    df[col] = df[col].astype(str).str.strip().str.capitalize()

scale_cols = ['ApplicantIncome', 'CoapplicantIncome', 'LoanAmount']

min_max_scaler = MinMaxScaler()
df[scale_cols] = min_max_scaler.fit_transform(df[scale_cols])

standard_scaler = StandardScaler()
df[['Credit_History']] = standard_scaler.fit_transform(df[['Credit_History']])

print("\nCleaned Data:")
display(df)

print("\nFinal Data Information:")
df.info()
```

OUTPUT:

```
Initial Data Overview:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11 entries, 0 to 10
Data columns (total 11 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Gender             10 non-null     object
 1   Married            11 non-null     object
 2   Dependents         11 non-null     object
 3   Education          11 non-null     object
 4   Self_Employed      11 non-null     object
 5   ApplicantIncome    11 non-null     int64
 6   CoapplicantIncome  11 non-null     int64
 7   LoanAmount         10 non-null     float64
 8   Loan_Amount_Term   11 non-null     int64
 9   Credit_History     10 non-null     float64
 10  Property_Area      11 non-null     object
dtypes: float64(2), int64(3), object(6)
memory usage: 1.1+ KB
None


Missing Values:
Gender               1
Married              0
Dependents           0
Education            0
Self_Employed        0
ApplicantIncome      0
CoapplicantIncome    0
LoanAmount           1
Loan_Amount_Term     0
Credit_History       1
Property_Area        0
dtype: int64

Duplicates Removed: 1
```

```
Cleaned Data:
/tmp/ipykernel_11634/1792679482.py:46: FutureWarning: Downcasting behavior in `replace` is deprec
  df['Dependents'] = df['Dependents'].replace('3+', 3)
```

	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	Loan
0	Male	Yes	1	Graduate	No	0.40	0.60	
1	Female	No	0	Graduate	No	0.00	0.00	
2	Female	Yes	2	Not graduate	Yes	0.80	0.80	
3	Female	Yes	1	Graduate	No	0.30	0.40	
4	Male	No	3	Graduate	Yes	1.00	1.00	
5	Female	Yes	0	Not graduate	No	0.10	0.20	
6	Male	No	2	Graduate	No	0.60	0.48	
7	Female	Yes	1	Graduate	Yes	0.20	0.00	
8	Male	Yes	0	Not graduate	No	0.50	0.72	
9	Female	No	3	Graduate	No	0.04	0.28	

```
Final Data Information:
<class 'pandas.core.frame.DataFrame'>
Index: 10 entries, 0 to 9
Data columns (total 11 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Gender             10 non-null     object
 1   Married            10 non-null     object
 2   Dependents         10 non-null     int64
 3   Education          10 non-null     object
 4   Self_Employed      10 non-null     object
 5   ApplicantIncome    10 non-null     float64
 6   CoapplicantIncome  10 non-null     float64
 7   LoanAmount         10 non-null     float64
 8   Loan_Amount_Term   10 non-null     int64
 9   Credit_History     10 non-null     float64
 10  Property_Area      10 non-null     object
dtypes: float64(4), int64(2), object(5)
memory usage: 960.0+ bytes
```